Least squares regression line

1 Any decimal between 0.8 and 0.99.

2

a $y = 19 - 2x$

b $y = -5 + 2x$

3 Weak and positive association.

4

a

i -0.85

ii Strong negative linear relationship

iii $y = -0.1x + 30.3$

b

i 0.75

ii Moderate positive linear relationship

iii $y = 5.8x - 99.0$

5

a $n = -2.50t + 20.94$

b $r = -0.89$

c As the temperature increases, the number of hot cocoa sold will decrease.

6

a $r = 0.86$

b Strong positive linear relationship

c $y = 0.89x + 2.78$

d $25.03^\circ\mathrm{C}$

e $32^\circ\mathrm{C}$, because the other two values are examples of extrapolation.

7

a $r = -0.908$

b Strong negative linear relationship

c $y = -0.06x + 6.37$

d $6.07\ \mathrm{kg}$

e Reliable due to interpolation and a strong correlation.

8

- **a** $r = 0.86$
- **b** Strong positive linear relationship
- **c** $y = 0.98x - 7.94$
- **d** $\$50.86$
- **e** Reliable due to interpolation and a strong correlation.

9

- **a** $r = -0.91$
- **b** Strong negative linear relationship
- **c** $y = -6.6x + 80.1$
- **d** 47.1
- **e** Reliable due to interpolation and a strong correlation.

10

- **a** $r = 0.67$
- **b** Moderate positive linear relationship
- **c** $y = 0.21x - 2.09$
- **d** 15 computers
- **e** Very unreliable due to extrapolation and a moderate to weak correlation.

11

- **a** $r = 0.79$
- **b** Strong positive linear relationship
- **c** $y = 0.025x - 550.108$
- **d** $\$1449.89$
- **e** $\$80\,000$ because the other two are examples of extrapolation.

12

- **a** $r = 0.997$
- **b** Strong positive linear relationship
- **c** $y = 0.85x + 1.28$
- **d** 52.28%
- **e** Reliable due to interpolation and a strong correlation.

13

a $r = -0.20$

b $(1, 1.28)$

c $r = 0.79$

d Strong positive linear relationship

e $y = 0.05x + 0.23$

f 1.23

g Despite a moderate correlation, unreliable due to extrapolation.

Calculating residuals

14

a

x	y	$\hat{y}$	Residual
10	29	25	4
14	29	37	-8
7	20	16	4
6	21	13	8
20	58	55	3
13	26	34	-8

b

x	y	$\hat{y}$	Residual
12.8	-32.7	-33.3	0.6
11.4	-30.3	-30.3	0
11.1	-29.5	-29.7	0.2
6.5	-21.5	-20.1	-1.4
10	-28.5	-27.4	-1.1

15

a

x	1	3	5	7	9
y	22.7	22.3	24.2	21.8	21.5
$\hat{y}$	25.2	23.4	21.6	19.8	18
Residuals	-2.5	-1.1	2.6	2	3.5

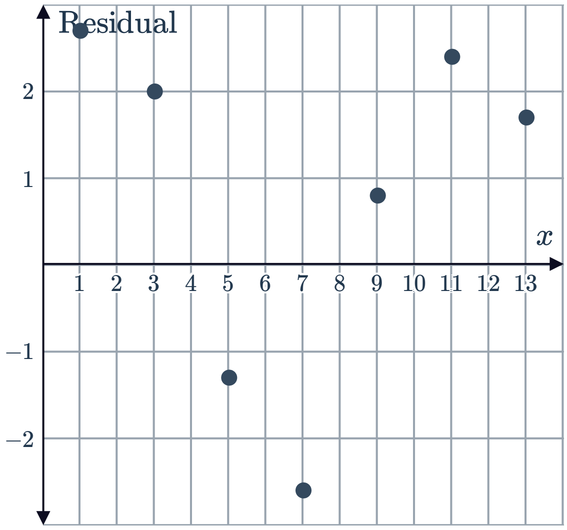
b

x	5	6	7	8	9
y	37.7	37.2	21.1	27.1	44
$\hat{y}$	28.9	30.4	31.9	33.4	34.9
Residuals	8.8	6.8	-10.8	-6.3	9.1

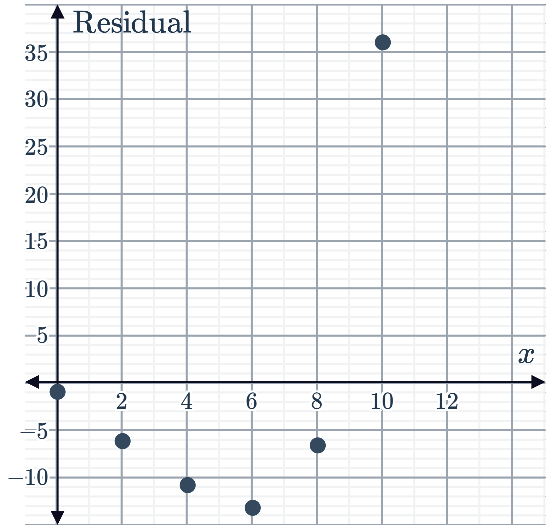
Residual plots

16

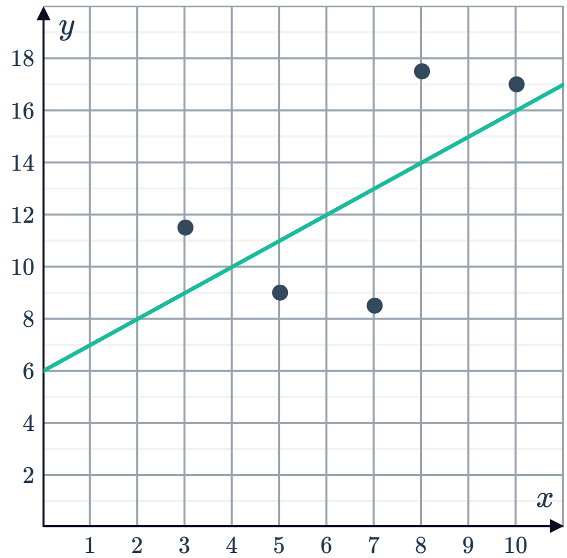
a



b



17



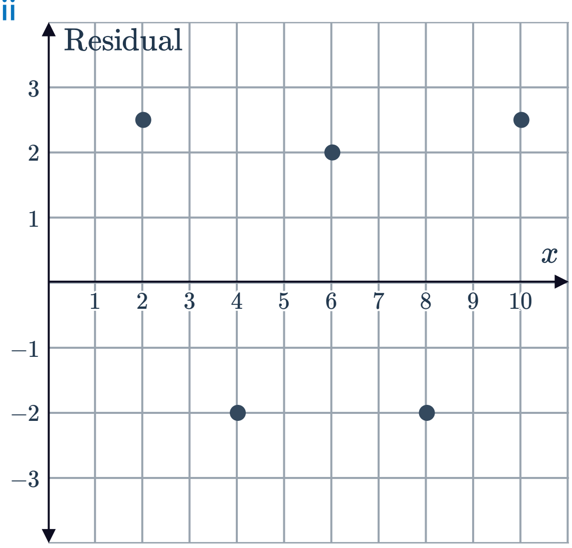
18

a

i

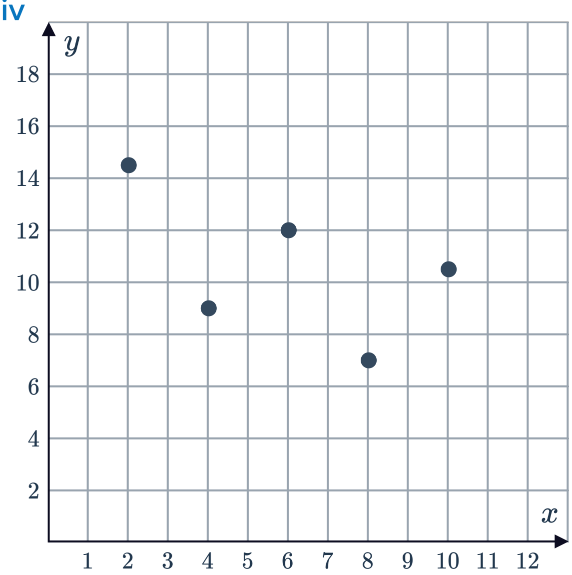
x	y	$\hat{y}$	Residuals
2	14.5	12	2.5
4	9	11	-2
6	12	10	2
8	7	9	-2
10	10.5	8	2.5

ii



iii Linear relationship

iv

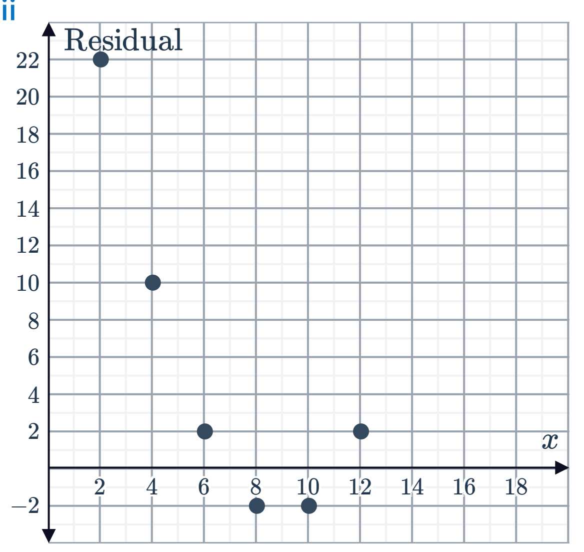


b

i

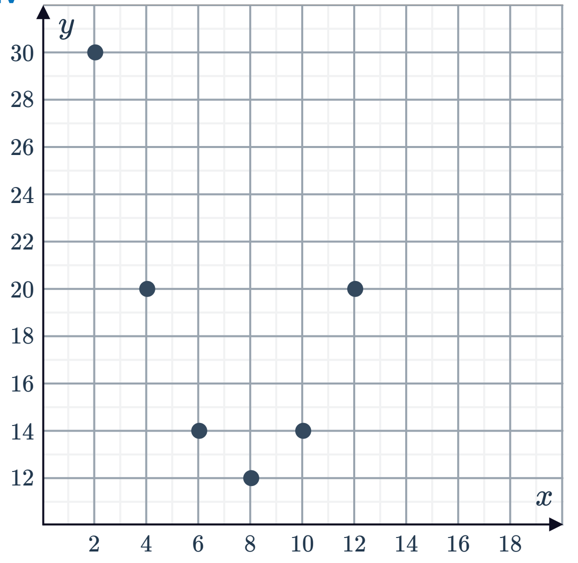
x	y	$\hat{y}$	Residuals
2	30	8	22
4	20	10	10
6	14	12	2
8	12	14	−2
10	14	16	−2
12	20	18	2

ii



iii Non-Linear relationship

iv



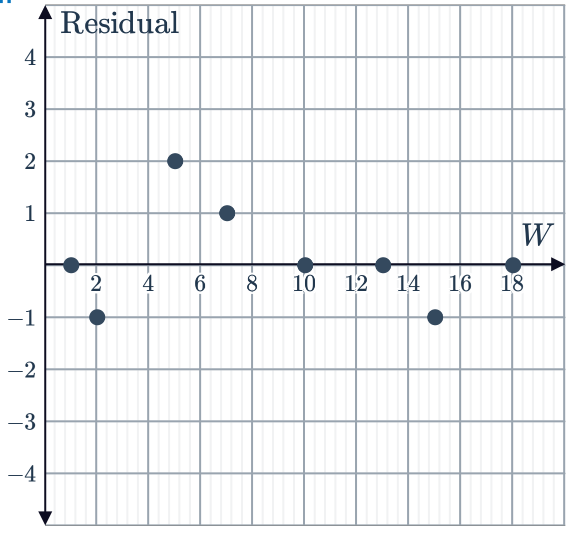
19

a

i

W	C	Model value	Residual
1	7	7	0
2	9	10	−1
5	21	19	2
7	26	25	1
10	34	34	0
13	43	43	0
15	48	49	−1
18	58	58	0

ii



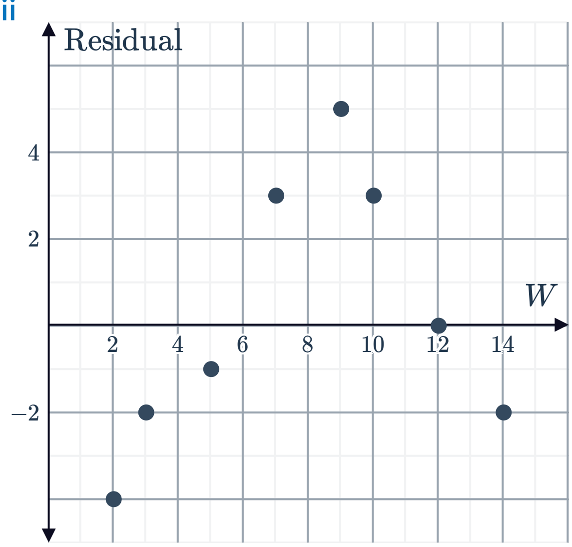
iii Yes, the model is a good fit.

b

i

W	R	Model value	Residual
2	7	11	−4
3	12	14	−2
5	19	20	−1
7	29	26	3
9	37	32	5
10	38	35	3
12	41	41	0
14	45	47	−2

ii



iii No, it is not a good fit.